KAMARAJUGADDA CHARAN SAI

FULL-STACK SOFTWARE ENGINEER & DIGITAL SOLUTIONS DEVELOPER

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PROFESSIONAL SUMMARY

Dynamic, results-driven Software Engineer and B.Tech CSE (Data Science) undergraduate with expertise across modern full-stack web architectures, Java desktop automation systems, and relational cloud database design. Experienced in building production web applications with Next.js/React, developing standalone Java software with vector PDF automation, and engineering secure Node.js backends with cryptographic hashing (PBKDF2), session management, and brute-force throttling. Passionate about delivering clean code, intuitive user experiences, and scalable solutions.

TECHNICAL EXPERTISE

Languages & Core:	`Java` `Python` `JavaScript (ES6+)` `TypeScript` `SQL` `HTML5 / CSS3`
Frontend & UI:	`React` `Next.js (App Router)` `Tailwind CSS` `Flutter & Dart` `Three.js` `Responsive Design`
Backend & Cloud:	`Node.js` `Express` `PostgreSQL` `Neon Cloud DB` `RESTful APIs` `Firebase`
Dev Tools & Platforms:	`Git & GitHub` `Render` `Netlify` `Linux` `Postman` `VS Code` `PDF Automation`

FEATURED PROJECTS

LETTERGEN PRO — Desktop Letter Generation & Document Automation Suite

2024 – 2026

`Java` `Java Swing` `PostgreSQL` `Neon Cloud` `Vector PDF Engine`

- Architected and implemented a standalone Java Swing desktop application designed for dynamic formal letter drafting, customized templating, and instant vector PDF document generation.
- Engineered two-way relational database synchronization with cloud-hosted PostgreSQL on Neon to guarantee centralized, persistent template and profile storage.
- Built robust input validation, document preview workflows, and clean local caching mechanisms for rapid document generation.

PROJEX — Engineering Student Career Readiness & Project Milestone Platform

2024 – Present

`Next.js` `React` `TypeScript` `Tailwind CSS` `PostgreSQL`

- Designed a modern full-stack web platform empowering engineering undergraduates with structured skill pathways, interactive curriculum roadmaps, and verified project milestone tracking.
- Built responsive client dashboards, interactive assessment modules, and type-safe API routes backed by relational PostgreSQL schemas.

MAMEKA MAHODAYAM — Digital Newspaper & Regional Publishing Web Portal

project-mm-1.onrender.com

`Next.js` `JavaScript` `Render` `Digital Publishing` `Cloud CDN`

- Engineered a high-traffic regional journalism platform tailored for Telugu vernacular typography, digital e-paper edition viewing, and dynamic editorial publishing workflows.
- Optimized server-side rendering and asset caching strategies to ensure fluid reading experiences across low-bandwidth mobile connections.

COMMERCIAL CLIENT SOLUTIONS — Scalable Business Portals & Digital Platforms

2023 – Present

`React` `Next.js` `Node.js` `Tailwind CSS` `Security & SEO`

- Engineered modern, responsive commercial portals for independent businesses with automated inquiry handling, spam honeypot filters, and sub-second page performance.

PROFESSIONAL EXPERIENCE

Independent Software Developer & Technical Consultant

2023 – Present

Freelance & Open-Source Projects • India

- Developed and deployed full-stack web applications and desktop utilities with persistent cloud storage, responsive UI, and custom administrative management dashboards.
- Implemented end-to-end security remediation: PBKDF2 password hashing with cryptographically secure salts, JWT authentication with session revocation via token versions, IP brute-force rate-limiting, and strict Content Security Policies (CSP).

EDUCATION

Bachelor of Technology (B.Tech) — Computer Science & Engineering

2022 – 2026

St. Ann's College of Engineering & Technology • Chirala, Andhra Pradesh, India

- **Specialization:** Data Science & Software Systems Engineering.
- **Key Coursework:** Data Structures & Algorithms (DSA), Database Management Systems (DBMS), Operating Systems, Computer Networks, Software Engineering, Object-Oriented Programming (Java/Python).